Interactive Healthy Blood

A pathway atlas from one gene module to a thousand donors

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1. What I seek to convey [1 pt]

Healthy whole blood is the most-measured human tissue in transcriptomics, yet every available dataset is visualised in isolation: Tabula Sapiens captures it at single-cell resolution, GTEx provides 803 cross-sectional bulk donors, GSE223613 follows 10 donors every three hours across a full day. Nobody has built a unified interactive view that lets a user zoom from the molecular to the cohort scale within a single tool.

I propose an **interactive atlas with five scales** – one gene cluster, one cell, one sample, one sample over time, many samples – with **pathway activity as the lens that connects all five**. A pathway like *interferon-alpha response* is meaningful at every scale: which genes make it up, which compartments of the cell those genes occupy, whether it is on in a given cell type, how enriched it is in one donor, whether it oscillates with time of day, how much cohort variance it drives. The glyph changes at every scale; the pathway label never does. That continuity is the educational and analytical payoff. The framework is tissue-agnostic – blood is the first instantiation; the same five-scale template extends naturally to liver, brain, and immune tissues.

Two concrete things the atlas will show:

(a) **Pathways are the cross-scale Rosetta stone.** The same pathway reads as a binary in a single cell, a continuous enrichment score in one bulk donor, an oscillating signal in a circadian time-course, and a principal axis of cohort variance – and all four readings agree on biology.

(b) **GTEx whole-blood variance is structured, not random.** A user interactively toggling the HSP-stress, circadian, and neutrophil-fraction pathway tokens will watch apparent “donor-to-donor noise” decompose into time-of-day, cell-composition, and genuine ex-vivo handling artefacts – structure that is invisible until you have the right lens.

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2. Visualisation idea [2 pts]

Hero, the persistent pathway constellation.

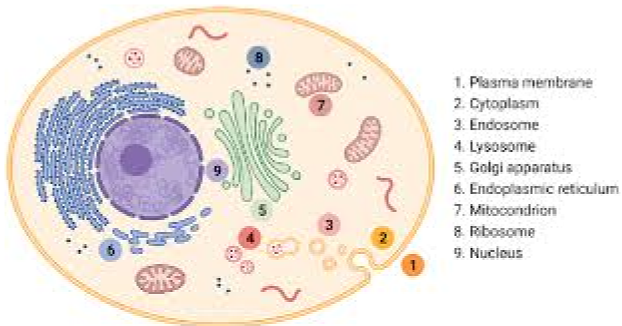
A floating panel of ~24 pathway-tokens (IFN-alpha, TCR-sig, OxPhos, HSP-stress, Circ-core, Neut-fraction, …) sits beside the main canvas at every scale. Each token’s fill encodes pathway activity score; its halo encodes statistical confidence. The constellation is the *lens*; the main canvas is the *specimen*. Clicking a token keeps it selected across all scale changes so the audience can watch the same pathway from gene module to cohort.

Five scales, five glyphs:

Scale	Dataset	Glyph	Pathway tokens
1 gene cluster	curated module (mito / ribo / HLA / histone)	co-expression network	module's enriched pathways
1 cell	Tabula Sapiens blood	**Cell anatomy diagram** + UMAP	binary on/off per compartment
1 sample	GTEx or GSE313156 W0	radial petal glyph	enrichment z-score
1 sample x 24h	GSE223613 (1 donor, 8 timepoints)	clock face + animated petal	per-timepoint score
N samples	GTEx (n=803)	Andrews curves + per-donor strip	cohort-variance fraction

Why start at gene cluster? Because a co-regulated module is the smallest pathway-like unit – a set of genes that *act as one thing*. The 37 mtDNA-encoded genes are the canonical example: a physically separate genome, co-translated by mitochondrial ribosomes, rising and falling together with oxidative-phosphorylation demand. Showing the internal structure of a module grounds the abstraction before zooming out.

The cell anatomy view is the novel centrepiece at the "1 cell" scale. Instead of a UMAP, the main canvas shows a schematic cross-section of a blood cell with labelled compartments: nucleus, mitochondria, ER/Golgi, cytoplasm, cell membrane, and – for immune cells – the immunological synapse / TCR complex. Every gene in the dataset is placed in its primary localisation (HPA subcellular atlas + UniProt topology). Clicking a compartment reveals the genes resident there with expression bars; clicking any gene opens a five-scale modal. Pathway tokens light the compartment their genes occupy, so the audience sees *where* a pathway lives inside the cell.



Cell anatomy view: compartments are clickable; selecting a pathway highlights the compartments its genes occupy.

Ancillaries. (i) A *gene breakdown* on pathway click: top-20 genes, per-scale distributions. (ii) A *scale-comparison strip* at the bottom: the selected pathway's reading at every scale on one line. For OxPhos: *37 genes, confined to mitochondria, on in 0.9 of one T cell, enriched in 0.7 of one donor, +/-0.2 over 24 h, 6 % of cohort variance.*

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3. How I will convey this story [2 pts]

Structure: Drill-Down. The five scales are zoom levels on the same biology, not a temporal narrative. Martini-Glass would impose a single linear order; Interactive Slideshow would silo each scale and break the unifying-pathway thread. Drill-Down matches the tool framing: it rewards the curious novice (zoom freely) and the analyst with a specific question (jump directly to cohort).

Storyboard.

1. **Landing.** A 5-tile row – one mini-canvas per scale, all showing the same default pathway (*OxPhos*). Caption: *"One pathway, five scales of healthy blood. Pick one."*
2. **Constellation banner** always visible. The audience sees what pathways exist before picking a scale.
3. **User clicks a scale.** Canvas opens with a 3-step guided intro (for *gene cluster*: "37 mtDNA genes -> edges = co-expression -> they rise and fall together"). Then free exploration.
4. **User clicks a pathway token.** It stays highlighted across scale changes; the bottom strip updates so all five readings align.
5. **User clicks a gene.** A 5-panel modal: co-expression neighbours, cell-type distribution, donor distribution, 24-h trace, cohort PC1 loading.

Encodings.

- *Position*: scale-conventional (network / UMAP+anatomy / petal / clock / Andrews); coherent within a scale.
- *Colour*: single diverging palette anchored at zero, used everywhere. Pathway category: secondary hue band (immune-blue / metabolic-green / stress-red / housekeeping-grey).
- *Animation*: 24 h scale only. Oscillating tokens pulse; stable ones stay still. Encoding, never decoration.
- *Persistent selection*: the only state that survives a scale change.

Sequence. No required order. Default left-to-right zoom-out suits a guided tour; analysts jump straight to cohort. Demo path: 37 mtDNA genes -> one T cell with OxPhos lit in the mitochondria compartment -> one donor enriched for those genes -> that donor across 24 h -> the cohort ranked by OxPhos score.

Context for the audience. Two sentences on the landing:

> *"Each panel shows healthy blood at a different scale of resolution. The tokens above are biological pathways, click one to follow it through every scale."*

Scoring details (binary sc fraction, ssGSEA, sliding-window ssGSEA, PC-loading x variance) live in tooltips, not in the narrative.

Transitions. Scale changes cross-fade in ~250 ms. The constellation never re-renders – it is the visual anchor that proves the instrument has not changed. The bottom strip animates the five pathway readings between scales in real time. *Datasets:* GTEx v11 whole blood (n=803), Tabula Sapiens blood (sc), GSE223613 (10 donors x 8 timepoints), GSE313156 W0 baselines (n=30), curated modules (mito, ribo, HLA, histone). *Compute budget:* < 2 h on a laptop.